

Nicholas Henry

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EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Mechanical Engineering and Computer Science

Sep 2025 - May 2029

Coursework: Diff Eq, Lin Alg, Discrete Structures, Program Design and Implementation, Statics

GPA: 3.79

Activities: Northeastern Electric Racing (FSAE), Students for the Exploration and Discovery of Space, Deans List

Studied a Semester Abroad at Queen's University Belfast in Northern Ireland

SKILLS

Languages: Python, C, Java, C#, JavaScript, SQL, HTML/CSS, TypeScript

Frameworks & Libraries: React, Next.js, Node.js, Tensorflow, NumPy, Flask, Dash, Tailwind

Tools & Platforms: Git, Docker, Github Actions, Redis, Supabase, Linux, ROS 2, Grafana, CI/CD, Claude Code, Raspberry Pi, Vercel, Kubernetes

Engineering Skills: SolidWorks, MATLAB, Project Management, Problem Solving, Cross-Functional Collaboration

Certifications & Training: IBM Artificial Intelligence Fundamentals, Microsoft C# Certification, GitHub Foundations

EXPERIENCE

Northeastern Electric Racing - Python, Dash, React, Docker, Git, Redis, Grafana

Boston, MA

Lead Software Engineer for Simulation Team

Jan 2026 – Present

- Develop a parametric multi-body vehicle simulation tool for NER mechanical engineers to test and examine how changes to geometry and other variables go on to affect the speed and times of the car on custom created tracks
- Utilize Python as the backend to do all the physics with Dash, and React for the frontend of the site to easily create all graphs and metrics making the UI easy for people to use effectively
- Conduct meetings, delegate work to members, code reviews, and research how to advance the physics engine

Students for the Exploration and Development of Space - Python, ROS 2, Git, Linux, Docker

Boston, MA

Software Engineer for Network Exploration Swarm Toolkit Team

Jan 2026 – Present

- Program on a project to create a swarm of C.O.B.R.A. (Crater Observing Bio-inspired Rolling Articulator) snake robots to explore and develop multi-agent collaboration in order to complete teamwork based exploration tasks
- Create the camera sensor ecosystem consisting of a simultaneous mapping program for multi-agent systems

SEONeeded - Wordpress, SEOMoz, Semrush

Miami, FL

Web Optimization Engineer

Oct 2025 – Aug 2026

- Perform client-facing tasks like identifying keywords, website maintenance, and strategies to target in order to drive more traffic for certain services increasing click-on rate and sales for 15 clients across the Greater South Florida region
- Adapted to the new wave of generative engine optimization space by learning what makes LLM models reference client's websites and how certain worded prompts get tailored responses, recommendations, and links to sources

Search Engine Optimization Intern

Jun 2025 – Sep 2025

- Drove a 54% increase in website traffic through targeted optimization by leveraging WordPress and keyword research

PROJECTS

SheepFinder - React, Next.js, Vercel, SQL, Git, Tailwind

May 2026 - Present

- Developed a Next.js and React web app based on time studying abroad in Northern Ireland designed to aid farmers and create an ecosystem where townspeople can report sightings of lost animals
- Spoke with farmers from Northern Ireland to gain an insight into what is the main issue and the best ways to solve them and built an app around that feedback like how sheep markings are designated across farms in different towns
- Have at least 20+ unique documented uses in Northern Ireland and heard from a farm of the benefits of the site

Portfolio Website - HTML/CSS, JavaScript, SQL, Supabase, GitHub Actions, Git

Mar 2025 - Present

- Built a site with HTML/CSS and Javascript hosted on Github Pages with a custom domain to display projects
- Applied SQL databases with Supabase to have a global comment section where visitors can write something
- Use GitHub Actions in order to autoupdate this resume on the site to match this document as it changes in Google Docs

Text Transcriber - Python, TensorFlow, Google APIs, Git

Jul 2025

- Designed and trained an AI model using TensorFlow on 100,000+ character images to recognize and convert handwriting into text and to then convert into audio with Google TTS as part of a 3-person team
- Tested with a group of visibly impaired people to see and hear writing and signage enabling them to be less dependant on caretakers in everyday life, with successful results in the limited cohort of 88% understanding the responses